

Chapter 5

Profiling of Anti-glycan IgA and IgG in Oral Fluid Using a Glycan Microarray

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Abstract

Secretory immunoglobulins (sIg), primarily IgA, control the microflora of the mucous membrane, protecting it from pathogens, as well as promoting its colonization by commensals. Knowledge of the specificity of sIgA (as well as sIgG) is necessary to understand their functions. We present here a protocol for using printed glycan array to profile the glycan specificity of sIg in oral fluid, with particular attention to sample preparation of this difficult-to-analyze material. The list of glycans in the array includes oligosaccharides typical of mucin *O*-chains, as well as bacterial polysaccharides, which are another full-fledged "player" in the sIg–mucin–microbiota system.

Key words Mucosal immunity, Oral fluid, Glycan microarray, PGA, Secretory IgA, Mucins, Anti-glycan antibodies, Microbiota

1 Introduction

Secretory immunoglobulins play important role in local communications between microbial community, host mucin interface, and immune system. The main classes of immunoglobulins in oral fluid are IgA and IgG (named further as sIgA and sIgG in order to differentiate them from blood antibodies). It's believed that normally sIgG enters the oral cavity from the peripheral blood circulation by passive transmucosal diffusion [1], although some of it is produced locally in the salivary glands or gingiva [2]. Its level significantly increased during infections and other diseases, as well as vaccination [3–5]. Once contacted a pathogen, sIgG eliminates it via complement activation and phagocytosis enhancement, i.e., plays a protective role [6–8].

sIgA is a dimeric molecule (higher oligomers also exist) fastened with joining (J) chain and decorated with secretory component (SC) [9]. sIgA has several functions. First of all these are

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neutralization of pathogenic microorganisms (viruses, bacteria, fungi) by the opsonization, blocking of attachment to the cell surface receptors, inhibition of pathogen mobility and their growth as well as preventing the access of antigen to immune system and implemented through the canonical interaction *antigen-F_{ab}* [8–10]. Noncanonical interactions occur through the binding of highly glycosylated SC with microbial lectins [11, 12]. IgA does not activate the complement system (as some IgG isotypes does) and even inhibits it [13], although activation of the lectin pathway by recruiting mannose-binding lectin has been described, when sIgA and lectin act synergistically [14].

sIgA promotes the colonization of the oral cavity by commensal microbiota using the same mechanisms as when interacting with pathogenic microorganisms, i.e., through F_{ab} and SC, but without the manifestation of cytotoxicity (noninflammatory pathway), maintaining the balance between the host and the microflora. These properties are assumed to be characteristic of naturally occurring sIgA, the antigen specificity of which is evolutionarily selected [15–17]. sIgA is especially effective in implementing the so-called "immune exclusion" of antigens and toxins when it forms a complex with mucin through cysteine residues [18], although a mechanism involving SC is also considered [19–21].

A significant proportion of microbial antigens are the glycans (oligo- and polysaccharides), available in isolated or synthetic form, and which are convenient for immobilization. Arrays of these glycans as well as glycans of mammalian origin (often in format of printed glycan array, PGA) are used in studies of glycan-binding proteins (CBPs), including human blood anti-glycan antibodies [22–25]. Oral fluid is one of the most difficult objects for studying immunoglobulins due to its viscosity, the presence of mucins of different nature and antigenicity, non-Ig CBP, enzymes, etc. To overcome the associated problems when analyzing secretory immunoglobulins, alternate methods are being sought, in particular, IgA is isolated from the pooled saliva samples or from the culture fluid of the mucosa-derived B cells [21, 26, 27]. Complete as possible knowledge about the carbohydrate targets of secretory Ig is necessary for, firstly, to study the balance between microorganisms, epithelium, secretory mucins and immunoglobulins, and secondly, to elucidate the reasons for changes in the reactivity of the mucous membrane when exposed to a pathogen or in autoimmune disorders. This knowledge is certainly necessary for the development of methods for diagnosing diseases caused by pathogens (or other microbes) and monitoring the effectiveness of nasal (and other) vaccines [28].

Here we propose a sample preparation protocol that includes [1] high-speed centrifugation and [2] filtration of the diluted supernatant through a 0.22 μm filter, which together removes most of the mucins and bacteria, and [3] lowering the incubation

temperature to inhibit bacterial enzyme activity. These measures result in a low background when determining the glycan binding profile of sIgG and sIgA in oral fluid.

2 Materials

Prepare all solutions in the bidistilled water and using the analytical grade reagents unless indicated otherwise. Prepare and store all reagents at +4 °C unless indicated otherwise. Perform all experiments with human fluids wearing laboratory gloves and glasses. Follow all local waste disposal regulations when disposing of waste materials.

2.1 Oral Fluid Sample Preparation

1. Frozen human oral fluid in 0.5 mL aliquots in 1.5 mL microcentrifuge tubes (whole oral fluid collected in the morning 1–2 h before breakfast and teeth cleaning after rinsing the mouth with a tap water). Store at −70 °C.
2. 0.22 μm, 13 mm polyvinylidene fluoride (PVDF) syringe filters (*see* **Note 1**).
3. Microtube centrifuge.
4. −70 °C freezer for storage of oral fluids samples.
5. 10 mM phosphate-buffered saline (PBS): dissolve 0.35 g NaH_2PO_4, 1.28 g Na_2HPO_4, and 8 g NaCl in approximately 900 mL of water, and then make up to 1 L with water. Check that the pH of the resulting solution is 7.4.

2.2 Microarray Binding Assays

1. Glycan microarrays (Array-700, GlycoNZ, Auckland, New Zealand) with barcodes, containing 480 oligo- and 295 polysaccharides (or similar microarray). The ligands description done in the Table 1. Store at +20 °C under vacuum.
2. Block buffer: dissolve 3.8 g of boric acid in 400 mL of water, add 1000 μL of ethanolamine and 800 μL of the Tween 20, mix well, and check that the pH of the resulting solution is 8.5 (*see* **Note 2**).
3. Sample buffer (PBS-0.1%T-1%BSA): 0.1% (v/v) Tween 20 and 1% of the bovine serum albumin, BSA, fraction V, in the PBS.
4. Washing buffer (PBS-0.05T): 0.05% (v/v) Tween 20 in PBS.
5. Paper towel or Kimwipes.
6. AlexaFluor647-labeled goat anti-human IgG, Fcγ-specific (for example, Jackson ImmunoResearch, West Grove, PA, USA), 1.5 mg/mL solution in PBS, 0.05% sodium azide, and 15 mg/mL BSA (as provided by the manufacturer).
7. Cy3-labeled goat anti-human IgA, α-chain specific (for example, Jackson ImmunoResearch, West Grove, PA, USA),

Table 1
The main groups of glycoligands in composition of microarray

Designation, systematic, or trivial name	
1. Mucin-type (core and terminal fragments) and related structures	2. Sialoglycans
3. Blood groups related structures (ABO, Lewis, P, Ii, etc.)	4. *N*-acetyllactosamine containing glycans (including glycoprotein *N*-chains)
5. Glucuronic acid (GlcA), GlcA-containing disaccharides, hyaluronic acids	6. Sulfated glycans
7. Mannose-containing oligosaccharides	8. Lactose derivatives
9. Tumor-associated carbohydrate antigens	10. Glycosphingolipids
11. Saccharides—fragments of bacterial and plant polysaccharides	12. Bacterial capsular and *O*-polysaccharides

1.5 mg/mL solution in PBS, 0.05% sodium azide, and 15 mg/mL BSA (as provided by the manufacturer).

8. Five-pcs microscope slide boxes.
9. +4 °C refrigerator.
10. Orbital shaker.
11. Orbital thermoshaker.
12. Vortex.
13. Mini-centrifuge for slides.
14. Microarray fluorescence scanner (for example, InnoScan 1100 AL, Innopsys, Carbonne, France, or another microarray scanner).
15. Incubation chamber with a lid.

2.3 Software

1. Fluorescence scanner software (Mapix 8.1.1 if Innoscan 1100AL is used or another software connected with a scanner).
2. TIFF image analysis software (for example, Scan Array Express 4.0, PerkinElmer, Waltham, MA, USA).
3. Microsoft Excel.

3 Methods

3.1 Oral Fluid Sample Preparation

1. Remove oral fluid aliquots from the −70 °C freezer and thaw on ice. We typically process up to eight samples at a time.
2. Once thawed centrifuge the sample (10,500×*g*, 5 min), aspirate 100 μL, and dilute up to 800 μL with the sample buffer. Mix well using vortex, and then centrifuge the diluted sample again (10,500×*g*, 5 min).

3. Take 800 μL of supernatant, filter it through the 0.22 μm, and make up to 1 mL with the sample buffer.

3.2 Glycan-Binding Profiling of Mucosal IgG and IgA

3.2.1 Microarray Assay

1. Put microarrays to the Couplin Jar (or plastic box for five-pcs microscope slides). Pour enough blocking buffer just to cover the full printed area and gently shake on orbital shaker (100 rpm) for 2 h (*see* **Note 3**).
2. Remove microarrays from the jar, and wash them dipping into the glass with the washing buffer 10 times. Collect excess buffer by touching corners of arrays to a paper towel.
3. Put the microarrays to the incubation chamber on a back side, so that the barcode is facing up and reads correctly (pour water on a bottom of the chamber for humidifying).
4. Add 900 μL of the sample on the array (one sample—one array) carefully spreading it to the surface using automatic pipette (*see* **Note 4**).
5. Close the lid of the chamber and incubate microarrays at +4 °C in a fridge under the slight shake (30 rpm) overnight.
6. Remove samples from the microarrays, and wash them dipping consequently into the two glasses with the washing buffer 10 times in each. Collect excess buffer by touching corners of arrays to a paper towel, and dry off the backside of the slide (*see* **Note 5**).
7. Dilute both the anti-IgG-Alexa647 and anti-IgA-Cy3 antibodies in the sample buffer to a final concentration of 1 μg/mL in one tube (*see* **Note 6**).
8. Put the microarrays to the incubation chamber, and add 900 μL of solutions containing both anti-IgG and anti-IgA antibodies.
9. Close the lid of the chamber, and incubate microarrays at +37 °C in the thermoshaker under the slight shake for 1 h.
10. Remove solution from the microarrays, and wash them dipping consequently into the two glasses with the washing buffer 10 times in each.
11. Finally wash microarrays in water, collect excess buffer by touching corners of arrays to a paper towel, and spin dry in mini-centrifuge.
12. Place microarrays to a five-pcs microscope slide box and keep in the dark until scanning (*see* **Note 7**).
13. Scanning procedure has been described in detail in several protocols such as [29] or [30]. In brief, scan microarrays using a microarray fluorescence scanner using two types of laser, 532 and 635 nm, for detection of IgA and IgG, correspondingly (*see* **Note 8**). In our laboratory we have Innoscan

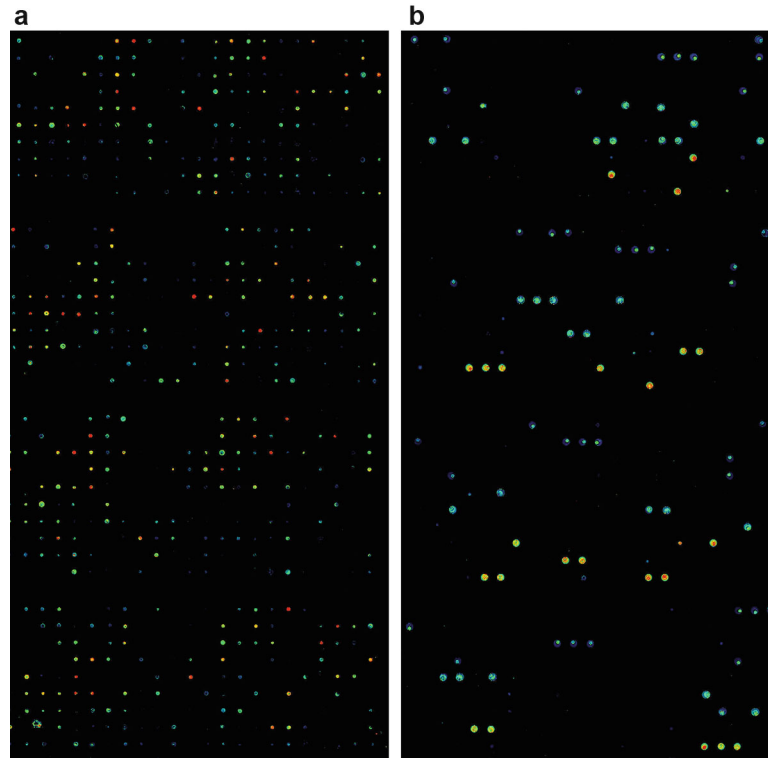


Fig. 1 Interaction of sIgA (**a**) and sIgG (**b**) from the oral fluid of a healthy donor with glycan array. sIgG visualized with Alexa647-labeled goat anti-human IgG and sIgA with Cy3-labeled goat anti human IgA. Images are presented in a rainbow color scale

1100AL, allowing to perform scanning on both wavelength simultaneously using Mapix software. Recommended scanning resolution is 10 μm. The examples of obtained images are seen in Fig. 1a, b.

3.2.2 Data Extraction and Analysis

Data extraction has been also described in detail in, e.g. [30]. In brief,

1. After scanning transfer obtained TIFF images to an image processing program. In our laboratory we use ScanArrayExpress 4.0. Other programs can also be used.
2. Using the GAL file (file containing information about the ligand identification numbers as well as their positions) convert image to the csv file (raw file) with digital data from the array.
3. Convert raw file to Excel xlsx file using any appropriate approach, for example, Excel macro. Obtained files from all tested samples can be collected in one file, sorted and graphed (*see* **Note 9**).

4 Notes

1. Other 0.22 μm filters with low protein binding can be used.
2. Use ethanolamine or boric acid to adjust pH in the Block buffer. Tween 20 is very viscose liquid; wait 2–3 s after aspiration and dispensing.
3. To avoid microdroplets of saliva released during breathing from contaminating the chip, we recommend working in a COVID-type mask.
4. Do not touch microarray surface by a pipette tip.
5. Do not allow microarray drying between incubation stages.
6. All needed solutions (except buffers) have to be freshly prepared (immediately before analysis).
7. After the arrays have been dried after the last incubation, they are ready to be immediately scanned. We recommend to scan right after the experiment or at least next day (stored at room temperature in the dark).
8. Scan settings are dependent on the sample and fluorescent label used. The microarray fluorescence scanner has a maximum relative fluorescence unit (RFU) of 65,535 for a saturated spot. It is best to have the strongest signals below this saturation for the most samples varying the PMT and/or laser power. Use the preview option to prevent having saturated signals.
9. Using the proposed protocol, an investigator takes into account the immunoglobulins in the liquid part of an oral fluid, which are not bound to mucins or bacteria removed during the sample preparation. It should be also noted that the total sIgA loss is less than 5%, in case of sIgG—about 10 % (unpublished data).

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